

Лабораторная работа №14

Администрирование локальных сетей

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Информация

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Вводная часть

- Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи

Основная часть

Настройка provider-sw-1

- Настройка интерфейсов коммутатора provider-sw-1

[illegible]

Настройка msk-donskaya-gw-1

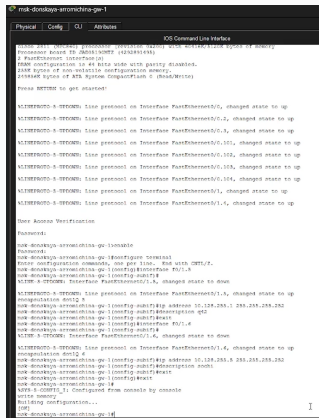


Рис. 1: Настройка интерфейсов маршрутизатора msk-donskaya-gw-1

Настройка msk-q42-gw-1

- Настройка интерфейсов маршрутизатора msk-q42-gw-1

```
msk-q42-w-1

Physics    Config    CU    Address

IOS Command Line Interface

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CLASS 211 (CMAC) processor (revision 0.0.0) with 4048K/32K6 bytes of memory
Processor Board ID: 20401A0072 (423293490)
CPU type: PowerPC (PowerPC)
CPU configuration: 4 or 8K bits with parity disabled.
32K bytes of non-volatile configuration memory.
Firmware type: 0 or 8K bytes (compactFlash 0 (Black/White))

Please RETURN to get started!

Key Access Verification

Password:

msk-q42-w-1enable
Password:
msk-q42-w-1configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-w-1(config)#interface P0/1.8
msk-q42-w-1(config-subif)#negotiation control 8
msk-q42-w-1(config-subif)#
msk-q42-w-1(config)#interface P0/1
msk-q42-w-1(config-if)#no shutdown

msk-q42-w-1(config)#exit
msk-q42-w-1#
msk-q42-w-1#show interface FastEthernet0/1, changed state to up
msk-q42-w-1#show interface FastEthernet0/1.8, changed state to up
msk-q42-w-1#show interface FastEthernet0/1.8, changed state to down
msk-q42-w-1#show interface FastEthernet0/1.8, changed state to up
msk-q42-w-1#show interface FastEthernet0/1.8, changed state to up
msk-q42-w-1(config)#interface P0/1.8
msk-q42-w-1(config-subif)#negotiation control 8
msk-q42-w-1(config-subif)#ip address 15.226.23.2 255.255.255.252
msk-q42-w-1(config-subif)#ip route 0.0.0.0 0.0.0.0
msk-q42-w-1(config-subif)#exit
msk-q42-w-1(config)#exit
msk-q42-w-1#
msk-q42-w-1#show controllers
Building configuration...
[OK]
msk-q42-w-1#
```

Настройка sch-sochi-sw-1

- Настройка интерфейсов коммутатора sch-sochi-sw-1

```
sch-sochi-sw-1
Physical Config CLI Advanced
IOS Command Line Interface

Class WS-C2850-24 (C2850) processor (revision C5) with 210592K bytes of memory.
Processor board ID FPC11228C
Running Standard Image
of FastEthernet/IEEE 802.3 interface(s)

4096K bytes of flash simulated non-volatile configuration memory.
Base ethernet MAC Address: 0000.0000.0000
Motherboard assembly number: 15-1761-05
Power supply part number: 14-0545-01
Motherboard serial number: FPC017610
Power supply serial number: 38N060312D
Model revision number: C5
Motherboard revision number: A0
Model number: WS-C2850-24
System serial number: FPC11228C

Class T000000000 Operating System Software
T00 (v0) C2850 Software (C2850-T4212-W), Version 12.1(20)EA4, RELEASE SOFTWARE(fx)
Copyright (c) 1986-2006 by Cisco Systems, Inc.
Compiled Wed 14-May-08 22:31 by jbarbera

Press RETURN to get started!

VLINE-0-CHANGED: Interface FastEthernet0/24, changed state to up
VLINEPROTO-0-DECODE: Line protocol on Interface FastEthernet0/24, changed state to up
VLINE-0-CHANGED: Interface FastEthernet0/1, changed state to up
VLINEPROTO-0-DECODE: Line protocol on Interface FastEthernet0/1, changed state to up

Done Access Verification
Password:
sch-sochi-sw-1>enable
sch-sochi-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-sw-1(config)#interface fa 0/24
sch-sochi-sw-1(config-if)#noarping noarping
sch-sochi-sw-1(config-if)#wait
sch-sochi-sw-1(config-if)#interface fa 0/24
sch-sochi-sw-1(config-if)#noarping noarping
sch-sochi-sw-1(config-if)#
VLINEPROTO-0-DECODE: Line protocol on Interface FastEthernet0/24, changed state to down
VLINEPROTO-0-DECODE: Line protocol on Interface FastEthernet0/24, changed state to up
VLINEPROTO-0-DECODE: Line protocol on Interface FastEthernet0/24, changed state to up
sch-sochi-sw-1(config)#vlan 4
sch-sochi-sw-1(config-vlan)#name mesh
sch-sochi-sw-1(config-vlan)#exit
sch-sochi-sw-1(config)#interface vlan 4
sch-sochi-sw-1(config-if)#
VLINE-0-CHANGED: Interface Vlan4, changed state to up
VLINEPROTO-0-DECODE: Line protocol on Interface Vlan4, changed state to up
sch-sochi-sw-1#
```


Настройка msk-q42-gw-1

```

root-q42-gw-1
Physics Config GUI Altrabax
KGS Command Line Interface

MTRM-8-CH00000: Terminate FastEthernet0/1, changed state to up
MTRM800-8-00000: Line protocol on Terminate FastEthernet0/1, changed state to up
MTRM-8-CH00000: Terminate FastEthernet0/1,8, changed state to down
MTRM800-8-00000: Line protocol on Terminate FastEthernet0/1,8, changed state to up
exit
now q42-gw-1#show ip interface FastEthernet 0/0/5
now q42-gw-1#show ip subif description description desc 2
now q42-gw-1#show ip subif address 10.129.1.1 255.255.255.0
now q42-gw-1#show ip subif description description desc2
now q42-gw-1#show ip subif ipsec
now q42-gw-1#show ipsec
now q42-gw-1#
MTRM-8-CH0001: Configured from console by console
write memory
Building configuration...
[OK]
now q42-gw-1#show interface tunnel
Error configuration commands, one per line. End with CMT/?.
now q42-gw-1#show ip interface FastEthernet 0/0/0
now q42-gw-1#show ip interface shutdown

now q42-gw-1#show ip f1
MTRM-8-CH0001: Terminate FastEthernet0/1, changed state to up
MTRM800-8-00000: Line protocol on Terminate FastEthernet0/1, changed state to up
exit
now q42-gw-1#show ip interface FastEthernet 0/0/201
now q42-gw-1#show ip subif
MTRM-8-CH00000: Terminate FastEthernet0/201, changed state to down
MTRM800-8-00000: Line protocol on Terminate FastEthernet0/201, changed state to up
now q42-gw-1#show ip subif address 10.129.1.1 255.255.255.0
now q42-gw-1#show ip subif description description ipsec-main
now q42-gw-1#show ip subif ipsec
now q42-gw-1#show ip interface f1/0
now q42-gw-1#show ip interface shutdown

now q42-gw-1#show ip f1
MTRM-8-CH0001: Terminate FastEthernet0/1, changed state to up
MTRM800-8-00000: Line protocol on Terminate FastEthernet0/1, changed state to up
exit
now q42-gw-1#show ip interface FastEthernet f1/0/201
now q42-gw-1#show ip subif f1
MTRM-8-CH00000: Terminate FastEthernet0/1,201, changed state to down
MTRM800-8-00000: Line protocol on Terminate FastEthernet0/1,201, changed state to up
now q42-gw-1#show ip subif address 10.129.1.1 255.255.255.0
now q42-gw-1#show ip subif description description ip2 management
now q42-gw-1#show ip subif ipsec
now q42-gw-1#show ipsec
now q42-gw-1#
MTRM-8-CH0001: Configured from console by console
write memory
Building configuration...
[OK]
now q42-gw-1#

```

Рис. 3: Настройка интерфейсов маршрутизатора msk-q42-gw-1

Настройка msk-hostel-gw-1

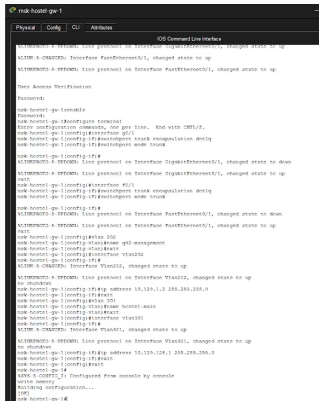


Рис. 5: Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-gw-1

Настройка msk-hostel-sw-1

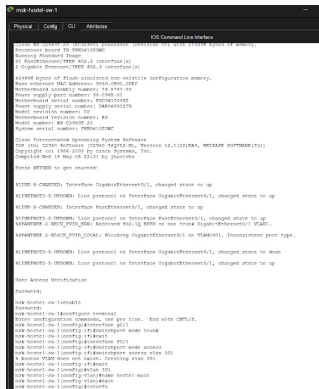
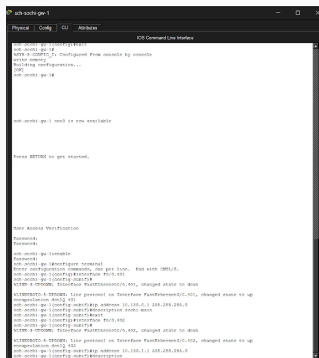


Рис. 6: Настройка интерфейсов коммутатора msk-hostel-sw-1

Настройка sch-sochi-gw-1



```
sch-sochi-gw-1
Physical Config CLI
IOS Command Line Interface

sch-sochi-gw-1#
sch-sochi-gw-1# display ipsec policy
Name: ipsec-policy
Protection mode: encrypt
Transform set: aes128-sha256
Sch-sochi-gw-1#

sch-sochi-gw-1# display ipsec policy
Name: ipsec-policy
Protection mode: encrypt
Transform set: aes128-sha256
Sch-sochi-gw-1#
```

Рис. 7: Настройка интерфейсов маршрутизатора sch-sochi-gw-1

Настройка msk-donskaya-gw-1

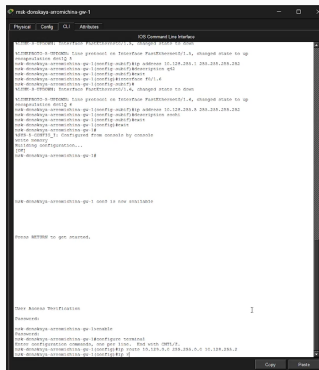


Рис. 9: Настройка маршрутизатора msk-donskaya-gw-1

Настройка msk-q42-gw-1

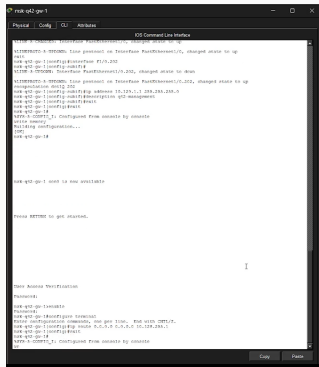


Рис. 10: Настройка маршрутизатора msk-q42-gw-1

Настройка sch-sochi-gw-1

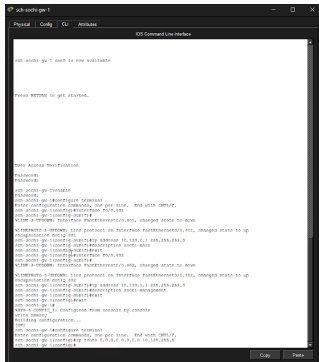


Рис. 11: Настройка маршрутизатора sch-sochi-gw-1

Настройка msk-q42-gw-1

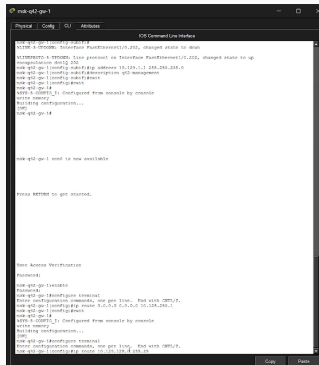


Рис. 12: Настройка маршрутизатора msk-q42-gw-1

Настройка msk-hostel-gw-1

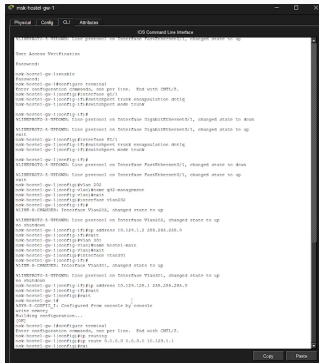


Рис. 13: Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-gw-1

Настройка msk-donskaya-gw-1

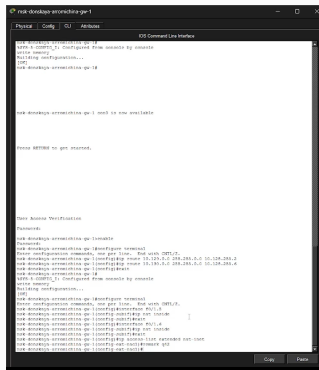


Рис. 14: Настройка NAT на маршрутизаторе msk-donskaya-gw-1

Вывод

- Мы смогли настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи